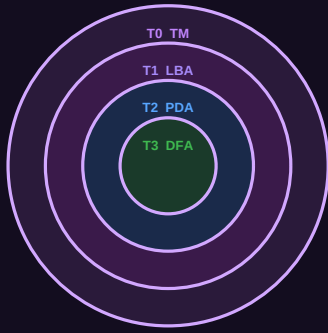
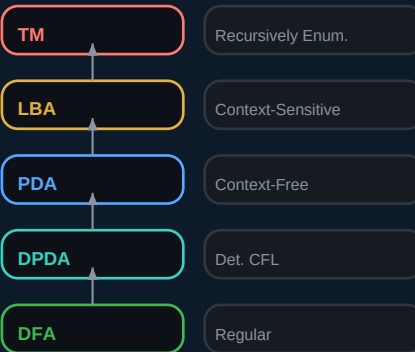


CHOMSKY HIERARCHY

Reg $\subset$ CFL $\subset$ CSL $\subset$ RE

AUTOMATA POWER



FORMAL DEFINITIONS

DFA: $M=(Q,\Sigma,\delta,q_{\text{start}},F)$ $\delta:Q\times\Sigma\rightarrow Q$ NFA: $M=(Q,\Sigma,\delta,q_{\text{start}},F)$ $\delta:Q\times\Sigma\rightarrow 2^Q$ PDA: $M=(Q,\Sigma,\Gamma,\delta,q_{\text{start}},Z_{\text{stack}},F)$ + StackTM: $M=(Q,\Sigma,\Gamma,\delta,q_{\text{start}},B,F)$ + Inf. TapeCFG: $G=(V,T,P,S)$ RE: $\emptyset \neq a (R_1+R_2)^*$ R_1R_2 R^*

DFA vs NFA

FINITE AUTOMATA

DFA

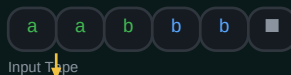
- $\delta:Q\times\Sigma\rightarrow Q$ (exactly 1)
- No ϵ -moves
- Deterministic

NFA

- $\delta\rightarrow 2^Q$ (0,1,many)
- ϵ -moves allowed
- Non-deterministic

NFA $\rightarrow$ DFA: Subset ConstructionEquiv. power, DFA may have 2^n states

PUSHDOWN AUTOMATON (PDA)



Finite Control

Stack

Z

DPDA

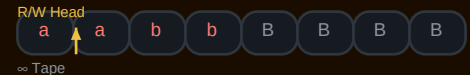
- Det. transitions
- Accepts DCFL

NPDA

- Non-det. guesses
- Accepts all CFL

DPDA $\neq$ NPDA e.g. $ww^R \in$ NPDA only

TURING MACHINE



Finite Control

 $\delta(q, a) \rightarrow (q', b, L/R)$

state read write move

Standard

Multi-Tape

NTM

UTM

All variants = Standard TM (same power)

REGULAR EXPRESSIONS & PUMPING LEMMA

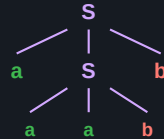
RE Operations

Concat ab $\{ab\}$ Star a^*

PUMPING LEMMA (Regular):

If L is regular, $\exists$ pumping length p : $\forall w \in L, |w| \geq p \exists w=xyz$ where: $|xy| \leq p$ $|y| > 0$ $xy^iz \in L \forall i \geq 0$ Use: Assume regular $\rightarrow$ find contradiction $\rightarrow$ NOT regular

CFG & DERIVATIONS

Parse Tree $S \rightarrow aSb|ab$ 

Yield: aabb

Leftmost:

 $S \rightarrow aSb \rightarrow a(ab)b \rightarrow aabb$
(expand leftmost NT first)

Rightmost:

 $S \rightarrow aSb \rightarrow aabb$
(expand rightmost NT first)

Ambiguous Grammar:

String has >1 parse treeCFG $\equiv$ PDA CFL accepted iff $\exists$ PDADPDA $\neq$ NPDA (ww^R needs NPDA)

DECIDABILITY

UNDECIDABLE:

No TM can always decide x

- Halting Problem (HALT_TM)

- Post Correspondence Problem

DECIDABLE (Recursive):

Halting Problem Proof:

Assume H solves HALT $\rightarrow$ Build $D(D)$ $D_{\text{halts}} \Leftrightarrow D_{\text{loops}} \rightarrow$ CONTRADICTION!

LANGUAGE OPERATIONS

Union $L \cup L'$: $a^* \cup b^* = \{a^* \text{ or } b^*\}$ Intersection $L \cap L'$: $\{a^m b^m\} \cap \{a^n b^n\}$ Difference $L - L'$: $a^* b^* - a^* = \{a^m b^m \mid m > 0\}$ Concat LL' : $\{a\}\{b\} = \{ab\}$ Kleene * L^* : $\epsilon \in L^*, \{ab\}^* = \{\epsilon, ab, abab\}$ Reversal L^R : $\{ab, ba\}^R = \{ba, ab\}$

MEALY VS MOORE

MEALY

 $\lambda: Q \times \Sigma \rightarrow \Delta$

Output on Transition

Fewer states

MOORE

 $\lambda: Q \rightarrow \Delta$

Output on State

More states

DFA MINIMIZATION:

- Remove unreachable states
- Partition: Final | Non-Final
- Refine until stable $\rightarrow$ minimal DFA

NFA $\rightarrow$ DFA SUBSET CONSTRUCTIONNFA: $q_0 \rightarrow \{q_0\}$ on 0, $\{q_1\}$ on 1 $q_1 \rightarrow \{q_1, q_2\}$ on 0, $\{q_1\}$ on 1 q_2 (final)

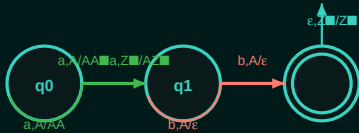
State	0	1
$\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\rightarrow \{q_0, q_1\}$	$\rightarrow \{q_0, q_1, q_2\}^*$	$\rightarrow \{q_0, q_1\}$
$^* \{q_0, q_1, q_2\}$	$\rightarrow \{q_0, q_1, q_2\}^*$	$\rightarrow \{q_0, q_1\}$

Final states: any DFA-state containing q_2 (NFA final)

CLOSURE PROPERTIES

Operation	Regular	CFL	CSL
Union	✓	✓	✓
Concat	✓	✓	✓
Kleene *	✓	✓	✓
Intersection	✓	✗	✓
Complement	✓	✗	✓
Difference	✓	✗	?
Reversal	✓	✓	✓

KEY THEOREMS: Arden's: $R=Q+R^P \rightarrow R=Q^P$ | Rice's: every non-trivial RE property is undecidable | Kleene: $\epsilon \in L^*$ always | Pumping(CFL): $w=uvxyz$ $|vx| > 0$ $uv^ixy \in L$

PDA for $L=\{a^n b^n\}$ 

Trace: aabb

Input	Stack	State
aabb	Z	q0
abb	AZ	q0
bb	AAZ	q0
b	AZ	q1

 $L=\{ww^R\}$: NPDA only (guesses midpoint)

NFA & ε-CLOSURE

ε-NFA: NFA with ε (empty string) transitions

ε-closure(q) = {q} ∪ {states reachable by ε}



ε-NFA → DFA Conversion:

- Start: ε-closure(q0)
- For each symbol a: $\delta(S,a) = \epsilon\text{-closure}(\cup \delta(q,a))$
- Mark state final if any final NFA state ∈ it
- Repeat until no new states

Every NFA is ε-NFA (ε-moves optional)

NFA: $|w| \geq 2$ over $\Sigma=\{0,1\}$ 

TM COMPUTATION & VARIANTS

Configuration: $\alpha q \beta$ α = left of head q = current state β = rightYields: $\alpha 1 q 1 \beta 1 \mid \alpha 2 q 2 \beta 2$ Language: $L(M) = \{w \mid M \text{ halts-accepts on } w\}$

UNIVERSAL TM (UTM):

UTM($\langle M \rangle, w$) simulates M on w

Basis of modern computers

Stores program + data on tape

Comparison

Feature	DFA	PDA	TM
Memory	States	Stack	∞ Tape
Language	Regular	CFL	RE
Write	No	No	Yes
Move L/R	No	No	Yes
Power	Least	Mid	Most

GRAMMAR CONVERSIONS & TYPES

Regular Grammar (Type-3):

Right-linear: $A \rightarrow aB \mid a$ Left-linear: $A \rightarrow Ba \mid a$

Both generate REGULAR languages

e.g. $S \rightarrow aA, A \rightarrow bA \mid b$ generates $L=\{ab^n\}$

Left Recursion Removal:

 $A \rightarrow A\alpha \mid \beta \rightarrow A \rightarrow \beta A' \quad A' \rightarrow \alpha A' \mid \epsilon$ e.g. $S \rightarrow Sa \mid b \rightarrow S \rightarrow bS' \quad S' \rightarrow aS' \mid \epsilon$ Application: $A=S, \alpha=a, \beta=b$

Chomsky Normal Form (CNF):

All productions: $A \rightarrow BC$ or $A \rightarrow a$

Every CFG can be converted to CNF

Used in CYK parsing algorithm

PUMPING LEMMA FOR CFL

Statement:

If L is CFL, $\exists$ pumping length p such that $\forall w \in L, |w| \geq p \rightarrow w=uvxyz$ where: $|vxy| \leq p \quad |vy| > 0 \quad uv^i xy^i z \in L \quad \forall i \geq 0$

Use: prove language is NOT context-free

Classic Non-Regular (use Pumping Reg.):

- $L=\{a^n b^n\} \rightarrow$ pump $y=a$ $\rightarrow a^{n+k} b^n \notin L$
- $L=\{a^n\} \rightarrow$ pump $y=a$ $\rightarrow p^2+k$ not perfect square
- $L=\{ww\} \rightarrow$ Not regular, IS context-free (NPDA)

Classic Non-CFL (use Pumping CFL):

 $L=\{a^n b^n c^n\} \rightarrow$ cannot pump both a and c simultaneously $L=\{a^n b^n c^n d^n\} \rightarrow$ Not CFL, needs LBA (CSL)

NFA → REGULAR EXPRESSION (Arden's Theorem)

Arden's Theorem:

If $R = Q + RP \quad (P \notin R) \rightarrow R = QP^*$ If $R = Q + PR \rightarrow R = P^*Q$ Example: $R = aR + b \rightarrow R = ba^*$

NFA → RE via State Equations:

- Write equations: $q_i = \Sigma$ (input-destination)
- Apply Arden's theorem to solve each equation
- Substitute back, simplify

 $q1 = (a+a(ab+b)^*b)^* \leftarrow$ example result

DFA EQUIVALENCE & MINIMIZATION

Equivalence Check:

- Pair initial states: (p, q)
- Check transitions: $(\delta(p,a), \delta(q,a))$ for each a
- Conflict if one is final, other is not $\rightarrow$ NOT equivalent
- All states matched $\rightarrow$ DFAs EQUIVALENT

Minimization (Table-Filling Method):

- Remove unreachable states
- Partition: Final (F) | Non-Final (Q-F)
- Mark pair (p,q) distinguishable if $\delta(p,a) \in F, \delta(q,a) \notin F$
- Repeat: mark (p,q) if $\exists a: (\delta(p,a), \delta(q,a))$ marked
- Merge all unmarked pairs $\rightarrow$ Minimized DFA

KEY IDENTITIES & FORMULAS

String Ops:

 $|ε|=0 \quad |w|=\text{length} \quad w^R=\text{reverse}$ $|w \cdot v| = |w| + |v|$ $\Sigma^* =$ all strings incl ϵ $\Sigma^+ = \Sigma^* - \{\epsilon\}$ $\epsilon \neq \emptyset \quad (\{\epsilon\} \neq \emptyset)$

RE Identities:

 $\emptyset^* = \epsilon^* = \epsilon$ $R + \emptyset = R \quad R \cdot \epsilon = R \quad R \cdot \emptyset = \emptyset$ $R^{**} = R^* \quad (R+S)^* = (R^*S^*)^*$ $R^* = \epsilon + R^+ \quad R^+ = R \cdot R^*$ $0^*1(0+10^*1)^* = (1+00^*1)(0+10^*1)^*$

Language Classes:

 $\text{Reg} \subset \text{DCFL} \subset \text{CFL} \subset \text{CSL} \subset \text{RE}$ $\text{DFA} \equiv \text{NFA} \equiv \epsilon\text{-NFA} \equiv \text{RE} \equiv \text{RG}$ $\text{CFG} \equiv \text{PDA (non-det)}$ DPDA $\neq$ NPDA $|\Sigma \leq n| = \Sigma \cdot \Sigma \cdot \Sigma \cdot \Sigma \quad |\Sigma|$

IMPORTANT LANGUAGES

 $a^n b^n$

CFL, not Regular

 $a^n b^n c^n$

CSL, not CFL

 ww^R

CFL (NPDA), not DCFL

 ww

CSL, not CFL

Palindromes

CFL

 $(ab)^*$

Regular

HALT_TM

RE, Undecidable

 $L=\{a^n\}$

Not Regular